

Developer & Operations Guide

D365FO Middleware Backend — Operations, Deployment & Azure DevOps Workflow

Target Audience: Developers, DevOps Engineers, System Administrators.

1. Quick Start & Local Setup

```
# Clone the repository
git clone https://MescoMBDTeam@dev.azure.com/MescoMBDTeam/D365FOMiddleware_Nestbackend/_git/D365FOMiddleware_Nestbackend
cd D365FOMiddleware_Nestbackend

# Install dependencies via pnpm
pnpm install

# Start local infrastructure (MongoDB 8 & Redis 7) via Docker
pnpm run docker:up

# Run application in development watch mode
pnpm run start:dev
```

2. API Endpoints Map (Swagger OpenAPI at /docs)

HTTP Method	API Route Endpoint	Purpose & Description	Required Role
POST	/api/v1/auth/login	Authenticate user & issue JWT Access/Refresh tokens	Public
POST	/api/v1/auth/refresh	Refresh access token using valid refresh token	Public
POST	/api/v1/data-batch/upload	Upload Excel file, parse, validate, and create batch	Operator / Admin
GET	/api/v1/data-batch	List all data batches with filter & pagination	Operator / Admin
GET	/api/v1/data-batch/:id	Get detailed batch status & line items	Operator / Admin
POST	/api/v1/data-batch/:id/process	Manually trigger queue processing for batch	Admin
POST	/api/v1/data-batch/:id/cancel	Cancel a pending batch	Admin
POST	/api/v1/master-data/sync	Trigger manual master data sync from D365FO	Admin
GET	/health	Application health probe (Mongo, Redis, D365FO status)	Public

3. Adding a New Entry Processor (Step-by-Step Guide)

To add a new Excel entry processor (e.g. for Customs Clearance AR):

1. **Define Raw Model:** Create raw data model in `src/modules/[domain]/models/`.
2. **Implement Processor Class:** Create class extending `EntryProcessorBase` in `src/modules/[domain]/processors/`.
3. **Register in Enum:** Add new type to `EntryProcessorTypes` enum in `src/modules/data-batch/enums/data-batch.enum.ts`.
4. **Register in Factory:** Add processor to `EntryProcessorFactory` in `src/modules/entry-processor/entry-processor.factory.ts`.
5. **Register in Module:** Add processor provider to `EntryProcessorsModule`.

4. Docker Production Build & Deployment

```
# Build production Docker image
docker build -t moatazali/middleware-app:1.0.2 .

# Push image to registry
docker push moatazali/middleware-app:1.0.2

# Remote deployment command
pnpm run docker:deploy
```

5. Git & Azure DevOps Integration Workflow

This project is hosted on Azure DevOps:

- **Repository URI:** `https://MescoMBDTeam@dev.azure.com/MescoMBDTeam/D365FOMiddleware_Nestbackend/_git/D365FOMiddleware_Nestbackend`
- **Main Branch:** `main`

How Documentation is Maintained in Azure DevOps:

- All Markdown documentation source files are stored in `docs/`.
- All compiled executive PDF manuals are stored in `docs_pdf/`.
- To push updates to Azure DevOps:

```
git add docs/ docs_pdf/
git commit -m "docs: update module documentation and PDF manuals"
git push origin main
```